Tu-Ruei (Jenson) Huang

Atlanta, GA 30332 | (404) 409-0580 | thuang407@gatech.edu | F-1 Visa

Education

Georgia Institute of Technology | Atlanta, GA *August 2025 – Present*

- Bachelor of Science in Electrical Engineering, GPA 4.00 Expected Graduation, May 2028
- Relevant Coursework: Digital System Design, Intro to Signal Processing, Differential Equations

Skills

Programming: Python, MATLAB (basic)
Hardware: logic analyzer, multimeter, soldering
Software: Altium Designer (basic)
Communication: presentations, research papers, teaching materials
Languages: English (native), Mandarin Chinese (native)

Experience

Uncommon Sense Lab, Georgia Tech | Atlanta, GA *September 2025 – Present*
Undergraduate Research Assistant

- Developing a low-cost breast cancer screening device that detects potential breast tissue abnormalities.
- Reduced PCB footprint by almost 40% through a layout redesign for mechanical constraints, followed by in-house fabrication via operating laser etching equipment and soldering components.
- Wired and interfaced PCBs with Arduino microcontrollers, ensuring reliability with logic analyzers for protocol debugging and multimeters for continuity testing.

Self-Employed | Hsinchu, Taiwan *July 2023 – August 2025*
Private Tutor

- Guided students to improve grades and master concepts in algebra, trigonometry, physics, and argumentative writing.
- Developed customized lesson plans to meet student needs, delivering instruction in 1-on-1, classroom, and virtual settings.

Projects

ACT Driving Simulator | Autonomous and Connected Transportation Lab, Georgia Tech *Fall 2025*
Vertically Integrated Projects Student

- Collaborated on a team of 8 people on leveraging LLMs for autonomous vehicle operations.
- Fine-tuned T5 model to improve vehicle analysis and compared predictions against baseline to validate improvements.
- Generated and plotted vehicle trajectory data using Waymo's open motion dataset.

Borohma | International Genetically Engineered Machine (iGEM) Competition *March – October 2024*
Dry Lab Sub-Team Lead

- Investigated the bioproduction of a camphor-based insect repellent to replace traditional compounds.
- Developed a transmembrane diffusion model to quantify compound release rates from minicell encapsulations.
- Engineered a Bass diffusion model using Python to forecast market adoption for a dual-function mosquito-repellent fragrance.
- Coded dry lab technical documentation and interactive visualizations for the team's informational website.

Activities

RoboJackets RoboNav | *Electrical Sub-Team Member* *September 2025 – Present*

- Applied skills in circuit analysis and soldering to build the electrical framework for a mini rover, simulating the University Rover Challenge.
- Successfully wired and interfaced a Bluetooth control module for the mini rover to enable wireless control during team challenges.

World Scholar's Cup Club, International Bilingual School at Hsinchu Science Park | *VP, Treasurer* *August 2021 – June 2025*

- Managed up to $80,000 USD worth of finances as the school's largest academic club with 100+ members.
- Organized logistics for international competitions in Malaysia and Thailand, and led delegation of 50+ students abroad.

Orchestra, International Bilingual School at Hsinchu Science Park | *President and Concertmaster* *August 2021 – June 2025*

- Hosted a charity concert with audience of 500+, raising $3,000 USD and 48 liters of cooking oil for dysfunctional families.
- Founded elementary school orchestra with 45 members, cultivating young musical talent and doubling overall membership.